



University of South Asia

Course Outline

Artificial Intelligence (CSC-3602)

Fall-2026

Instructor	Mr. Muhammad Waqas Ishaq
Office No.	C-21
Office Hours	08:00 AM – 04:00 PM (Mon-Fri)
Email	Waqas.ishaq@usa.edu.pk
Course URL (If any)	

Course Basics				
Credit Hours	3(2-3)			
LECTURES	Lectures per week	Two (2)	Duration	60 minutes each
LAB	Lectures per week	Two (2)	Duration	90 minutes each
Venue	G-7			
Visiting Hours	Duration	3 Hours		
TUTORIALS	Sessions per week	One (1)	Duration	40 minutes
Venue	G-7			

Course Distribution	
Core	Yes (Mandatory Major course for BS Computer Science as per HEC 2025 curriculum)
Elective	No
General	No
Prerequisite (s)	Data Structures and Algorithms

Course Description
Artificial Intelligence introduces the foundations of intelligent problem solving and decision making. The course covers intelligent agents, search, knowledge representation, reasoning, uncertainty, and selected learning approaches. It also provides introductory exposure to machine learning, deep learning, NLP, computer vision, generative AI, and agentic AI, with emphasis on real-world applications, ethics, and limitations.



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Course Contents

Introduction to AI; classical and modern AI; intelligent agents and PEAS; problem formulation; uninformed, informed and local search; constraint satisfaction; adversarial search; knowledge representation; propositional and first-order logic; reasoning and expert systems; uncertainty, Bayesian reasoning and fuzzy logic; planning; machine learning and reinforcement learning foundations; introductory deep learning, NLP and computer vision; generative AI and agentic AI; AI ethics, bias, fairness, privacy, safety, limitations and real-world

Graduate Attributes (GA)

- GA-1: Academic Education: To prepare graduates as computing professionals.
- GA-2: Knowledge for Solving Computing Problems: Apply knowledge of computing fundamentals, specialization, and relevant domain knowledge to formulate and solve computing problems.
- GA-3 Problem Analysis: Identify, formulate, research, and solve complex computing problems using fundamental principles of computing sciences and relevant disciplines.
- GA-4: Design/Development of Solutions: Design and evaluate solutions for complex computing problems that meet specified needs with appropriate societal considerations.
- GA-5: Modern Tool Usage: Create, select, adapt, and apply appropriate techniques, resources, and modern computing tools with an understanding of their limitations.

GA Mapping with CLOs

GAs\ CLOs	Bloom Taxonomy	GA-1	GA-2	GA-3	GA-4	GA-5	GA-6	GA-7	GA-8	GA-9	GA-10
CLO 1: Describe core AI concepts including search algorithms, knowledge representation,	C2 (Understand)	✓	✓								



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and reasoning.											
CLO 2: Implement basic AI algorithms for problem-solving and decision-making.	C3 (Apply)		✓			✓					
CLO 3: Analyze the applications of AI in real-world scenarios.	C4 (Analyze)			✓	✓						
CLO 4: Discuss ethical considerations and limitations of AI systems.	C2 (Understand)								✓	✓	
CLO 5: Develop simple AI models using appropriate tools and frameworks.	C6 (Create)				✓	✓					

Grading Scheme

Following is the criteria for the distribution of marks to evaluate final grade in a semester.

Quizzes	Assignments	Mid Term	Lab Marks	Final Exam	Total
7%	8%	25%	25%	35%	100%

Grading Breakup and Policy			
Assessments	Description	Weight (%)	CLOs
Quizzes	There will be 4 quizzes covering theoretical concepts and AI problem-solving.	7 %	CLO-1, 2, 3, 4
Assignments	There will be 4 assignments based on AI concepts, reasoning, case studies, and problem-solving.	8 %	CLO-1, 2, 3, 4
Lab/Project/ Presentation	A small AI project/presentation will be assigned in the last month of the semester.	25 %	CLO-2, 3, 4, 5



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Mid Term Exam	Mid Term will cover the partial Syllabus. It will be conducted in two parts. Part one will include MCQs. Part two will include Subjective Questions. Duration: 60~90 mins Expected Date: Mid-term: Expected in the 8th week Exam Specifications: Closed book, No calculator. No cell phones	25 %	CLO-1, 2
Final Exam	Final Term exam will cover the whole course content. It will be conducted in two parts. Part one will include MCQs. Part two will include Subjective Questions. Duration: 120~180 mins Expected Date: Final-term: Expected in the 16th week Exam Specifications: Closed book, No calculator. No cell phones	35 %	CLO-1, 2, 3, 4



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Class Policy

1. **Be On Time:** You need to be in the class at the assigned time. After 10 minutes past the assigned time, you will be marked absent.
2. **Mobile Policy: TURN OFF YOUR MOBILE PHONE!** It is unprofessional to be texting or otherwise.
3. **Email Policy: READ YOUR EMAILS!** You are responsible if you miss a deadline because you did not read your email. Participants should check their university emails accounts regularly and respond accordingly.
4. **Class Attendance Policy:** A minimum of 80% attendance is required for a participant to be eligible to sit in the final examination. Participants with less than 80% of attendance in a course will be given grade 'F' (Fail) and will not be allowed to take end term exams.
5. **Google Classroom:** Participants should regularly visit the course contents on Google Classroom, and fully benefit from its capabilities.
6. **Harassment Policy:** Sexual or any other harassment is prohibited and is constituted as punishable offence. Sexual or any other harassment of any participant will not be tolerated. All actions categorized as sexual or any other harassment when done physically or verbally would also be considered as sexual harassment when done using electronic media such as computers, mobiles, internet, emails etc.
7. **Use of Unfair Means/Honesty Policy:** Any participant found using unfair means or assisting another participant during a class test/quiz, assignments or examination would be liable to disciplinary action.
8. **Plagiarism Policy:** Plagiarism will not be tolerated and will result in action by the Discipline Committee.
9. **Communication of Results:** The results of quizzes, midterms and assignments are communicated to the participants during the semester and answer books are returned to them. It is the responsibility of the course instructor to keep the participants informed about his/her progress during the semester. The course instructor will inform a participant at least one week before the final examination related to his or her performance in the course.
10. **AI Usage Policy:** Students can use AI tools like ChatGPT for learning and simulation. Students must verify results and should not copy directly without understanding.



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Weekly Lecture Schedule

Weeks	Lectures	Topics	Assignment/Applications (AI-Based Activity)	AI Tools	CLO Covered	Bloom's Taxonomy Level(s)
1	1	Introduction to Artificial Intelligence: definition, goals, history and evolution	Class discussion: identify AI and non-AI systems from everyday examples.	ChatGPT, Gemini	CLO-01	Understand
	2	AI Landscape: classical AI, modern AI and major branches	Concept map: relate ML, NLP, vision, robotics and generative AI to AI.	ChatGPT, Gemini	CLO-01	Understand
2	3	Intelligent Agents: agents, environments, rationality, sensors and actuators	Case analysis: identify agent, environment, sensors and actuators in a real system.	ChatGPT	CLO-01	Understand
	4	PEAS, task environments and types of intelligent agents	PEAS activity for a self-driving car, medical assistant or vacuum agent.	ChatGPT	CLO-01	Understand
3	5	AI Problem Formulation: states, actions, goal test and path cost	Paper exercise: formulate a route-finding problem as a state-space problem.	ChatGPT	CLO-01, CLO-02	Understand, Apply
	6	Uninformed Search: BFS, DFS and Uniform-Cost Search	Trace search trees and compare completeness, optimality and complexity.	ChatGPT	CLO-02	Apply
4	7	Informed Search: heuristics and Greedy Best-First Search	Compare heuristic choices for a simple path-finding	ChatGPT	CLO-02	Apply



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			scenario.			
	8	A* Search: evaluation function, admissibility and consistency	Paper exercise: trace A* and explain how heuristics affect search.	ChatGPT	CLO-02	Apply
5	9	Local Search: Hill Climbing and Simulated Annealing	Case study: compare local-search strategies for an optimization problem.	ChatGPT	CLO-02, CLO-03	Apply, Analyze
	10	Constraint Satisfaction Problems: variables, domains and constraints	Model map-coloring or N-Queens as a CSP.	ChatGPT	CLO-02	Apply
6	11	Adversarial Search: game trees and Minimax	Trace a small game tree and determine the rational move.	ChatGPT	CLO-02	Apply
	12	Alpha-Beta Pruning and decision making in games	Compare Minimax with Alpha-Beta pruning on a sample tree.	ChatGPT	CLO-02, CLO-03	Apply, Analyze
7	13	Knowledge Representation: symbols, rules, semantic networks and frames	Represent a small real-world domain using rules or a semantic network.	ChatGPT	CLO-01	Understand



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	14	Propositional Logic: syntax, semantics and logical inference	Translate simple statements into propositional logic and evaluate them.	ChatGPT	CLO-01, CLO-02	Understand, Apply
8	15	First-Order Logic: predicates, variables and quantifiers	Translate natural-language statements into first-order logic.	ChatGPT	CLO-01, CLO-02	Understand, Apply
	16	Reasoning and Inference: unification, chaining and resolution	Compare forward chaining, backward chaining and resolution using examples.	ChatGPT	CLO-01, CLO-02	Understand, Apply
Mid Term Exam					CLO-01, CLO-02	Understand, Apply
9	17	Expert Systems: knowledge base, inference engine and rule-based reasoning	Case study: analyze the components of a medical or troubleshooting expert system.	ChatGPT	CLO-01, CLO-03	Understand, Analyze
	18	Uncertainty in AI: probability, Bayes theorem and Bayesian reasoning	Decision scenario: update belief using simple Bayesian	ChatGPT	CLO-02, CLO-03	Apply, Analyze



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			reasoning.			
10	19	Fuzzy Logic: fuzzy sets, membership functions and rules	Design a conceptual fuzzy rule set for temperature or speed control.	ChatGPT	CLO-02	Apply
	20	Planning and Decision Making in AI: goals, actions and plan formation	Case study: outline a simple goal-based plan and analyze alternatives.	ChatGPT	CLO-02, CLO-03	Apply, Analyze
11	21	Machine Learning Foundations: supervised, unsupervised and reinforcement learning	Classification activity: match real-world problems with suitable learning paradigms.	ChatGPT, Gemini	CLO-03	Analyze
	22	Supervised Learning Basics: classification, regression, training and evaluation	Case analysis: select a suitable supervised approach and evaluation measure.	ChatGPT	CLO-03	Analyze
12	23	Unsupervised Learning Basics: clustering and pattern discovery	Analyze where clustering can be useful in customer, text or image data.	ChatGPT	CLO-03	Analyze
	24	Reinforcement Learning: agent, state, action, reward and policy	Analyze a simple reward structure for a game or navigation agent.	ChatGPT	CLO-03	Analyze



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13	25	Deep Learning Basics: neural networks and representation learning	Concept activity: compare traditional ML with deep learning at a high level.	ChatGPT, Gemini	CLO-03	Analyze
	26	Natural Language Processing: text understanding and common applications	Case study: analyze sentiment analysis, translation and chatbot applications.	ChatGPT, Gemini	CLO-03	Analyze
14	27	Computer Vision: image understanding and common applications	Case study: analyze classification, detection and segmentation applications.	ChatGPT, Gemini	CLO-03	Analyze
	28	Generative AI: LLMs, multimodal generation, prompting and limitations	Compare generative and predictive AI using real-world examples.	ChatGPT, Gemini	CLO-03, CLO-04	Analyze, Understand
15	29	Agentic AI: autonomous agents, planning, memory, tools and workflows	Scenario analysis: distinguish a chatbot from an AI agent performing multi-step tasks.	ChatGPT, Gemini	CLO-03	Analyze



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	30	AI Ethics and Limitations: bias, fairness, privacy, safety and societal impact	Structured discussion: evaluate ethical risks and limitations of an AI system.	ChatGPT	CLO-04	Understand
16	31	Final Project Presentation: AI problem, approach, results and limitations	Student project presentations and feedback.	N/A	CLO-02, CLO-03, CLO-04, CLO-05	Apply, Analyze, Create
	32	Final Project Presentation and Course Synthesis	Student project presentations, feedback and integration of core AI concepts.	N/A	CLO-01, CLO-02, CLO-03, CLO-04, CLO-05	Understand, Apply, Analyze, Create
Final Term Exam					CLO-01, CLO-02, CLO-03, CLO-04	Understand, Apply, Analyze



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Project Details

Project Objective: The final project aims to apply fundamental AI concepts to a small real-world problem. Students will select an appropriate AI approach and develop a simple model or prototype while considering evaluation, limitations, and responsible AI practices.

Project Deliverables:

1. **Problem Statement:** Define a clear real-world problem, its objective, expected input/output, and the role of AI in addressing it.
2. **AI Method Design:** Select an appropriate AI approach such as search, rule-based reasoning, or a simple learning method, and justify the choice.
3. **Model/Prototype Development:** Develop a simple AI model or prototype using appropriate tools/frameworks and document the major steps and assumptions.
4. **Testing and Evaluation:** Evaluate the system outputs using suitable criteria, analyze errors, and identify technical and practical limitations.

Presentation: Present the problem, AI approach, results, limitations, ethical considerations, and potential real-world applications.

Recommended AI Tools: Python, Google Colab, scikit-learn, ChatGPT/Gemini, and other instructor-approved tools as appropriate.

Industry Tool Exposure: Students will receive conceptual exposure to modern AI workflows including model evaluation, responsible AI, version control, API-based AI services, and basic deployment considerations.

Responsible AI Review: Identify potential bias, privacy, safety, and misuse risks, and propose basic safeguards.



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Recommended Books

Text Books for the Course:

1. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 3rd edition, Prentice Hall, Inc., 2010.
2. Hart, P.E., Stork, D.G. and Duda, R.O., 2001. Pattern Classification. John Wiley & Sons.
3. Luger, G.F. and Stubblefield, W.A., 2009. AI Algorithms, Data Structures, and Idioms in Prolog, Lisp, and Java. Pearson Addison-Wesley.

Reference Books for the Course:

1. Norvig, P., Paradigms of Artificial Intelligence Programming: Case Studies in Common Lisp, Morgan Kaufmann, 1992.
2. Severance, C.R., Python for Everybody: Exploring Data Using Python 3, CreateSpace Independent Publishing Platform, 2016.
3. Miller, B.N., Ranum, D.L. & Anderson, J., Python Programming in Context, Jones & Bartlett Publishers, 2019.
4. Joshi, P., Artificial Intelligence with Python, Packt Publishing Ltd., 2017.